Diffusion Seminar (Week9)

Diffusion Language Models 2

Woojin Kim

AIDAS LAB
Seoul National University
wjk9904@snu.ac.kr

AIDAS Lab

Table of contents

1 Recap: Discreteness of Text

2 Discrete Diffusion Modeling

3 Masked Diffusion Language Models

4 Unmasking & Token Ordering

5 Next week

Recap: Discreteness of Text



Why Diffusion Doesn’t Directly Work on Text

- Unlike images, text is made of discrete tokens
  -> we can’t smoothly add noise like we do with pixels.

For text tokens

Text	Tokens
The cat sits on the mat	0, 40, 43, 13, 2, 67
↓	
? ? ? ? ? ?	0.5, 40.71, 42.1, 13.4, 1.8, 66.4
↓ Rounding?	
No image why on the love	1, 41, 42, 13, 2, 66

Fragility of Semantics

- In images, a small perturbation in a single pixel (e.g., brown → red) is less noticeable and rarely changes the overall meaning.

- In text, even a single token change (e.g., “a” → “b” or “cat” → “car”) can completely alter meaning and grammaticality.

“My cat ate a fish” → “My car ate b fish”

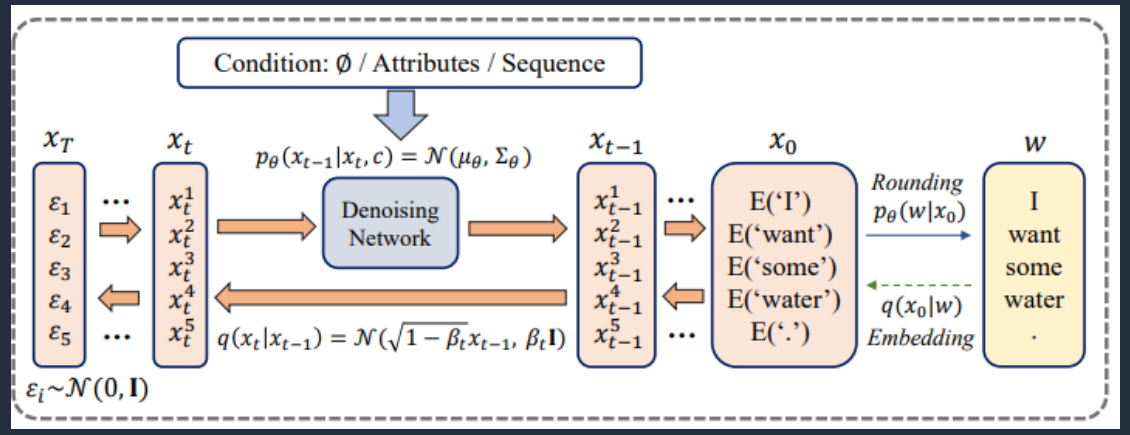
- Thus, it’s harder for diffusion in text to ensure that noisy intermediate states map back to fluent, coherent sentences.

Dealing with Discreteness

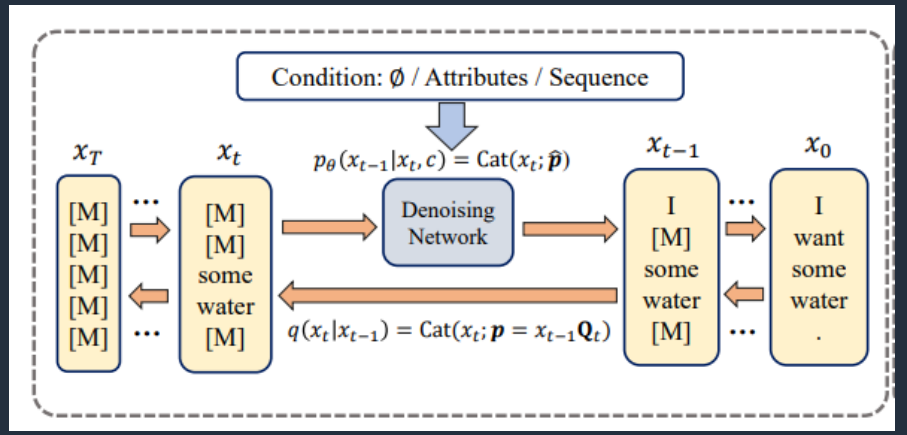
- To apply diffusion to language, we need to bridge the gap between discrete tokens and the continuous noise-based diffusion framework.

Two main paradigms

Continuous Diffusion



Discrete Diffusion



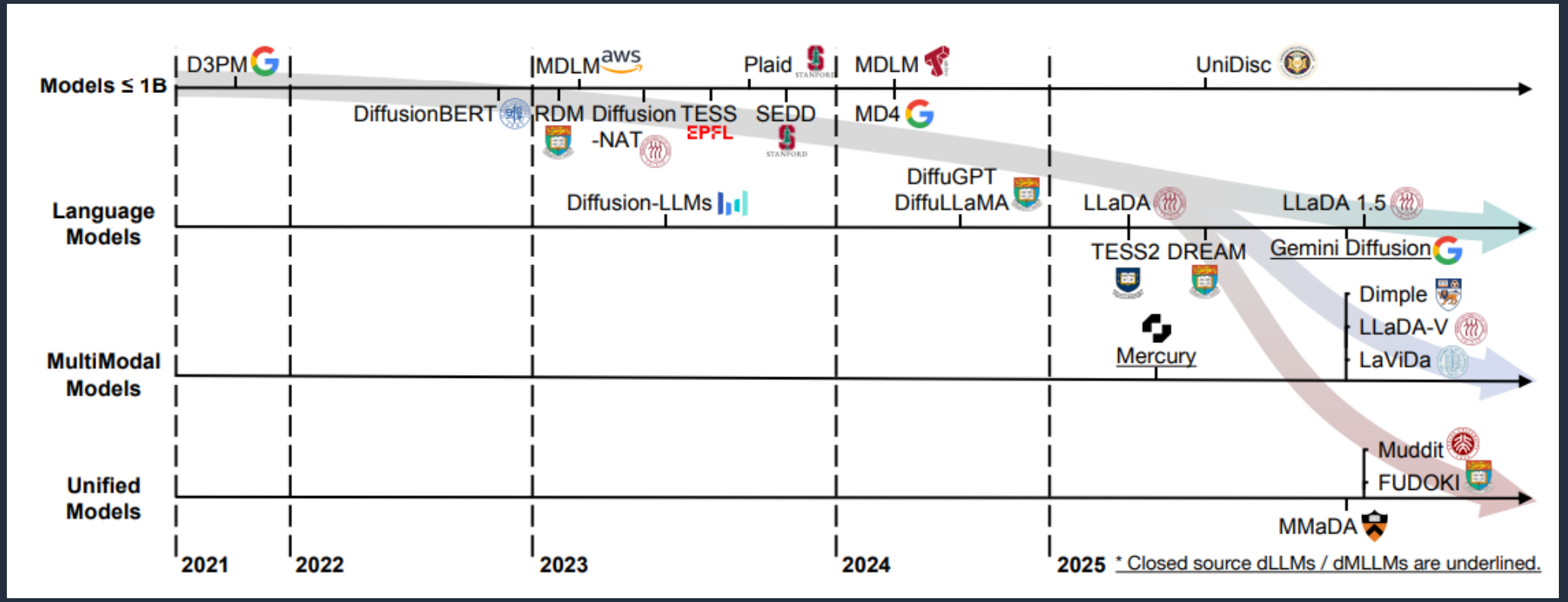
https://www.ijcai.org/proceedings/2023/0750.pdf

Continuous VS. Discrete Diffusion

	Continuous Diffusion LM	Discrete Diffusion LM
Space	Embedding, Latent	Token
Noising process	Continuous noising	Masking, Replacing
Pros	• Faster single-step operations (low-dim) • Classifier guidance • Image-based techniques are applicable	• Works directly on tokens • Easier alignment with existing models • Simplified training objective
Cons	• Training instability • Scaling to large corpora remains unproven • Requires encoder/decoder modules	• Classifier guidance is limited • Transition itself is not differentiable

Discrete Diffusion Modeling

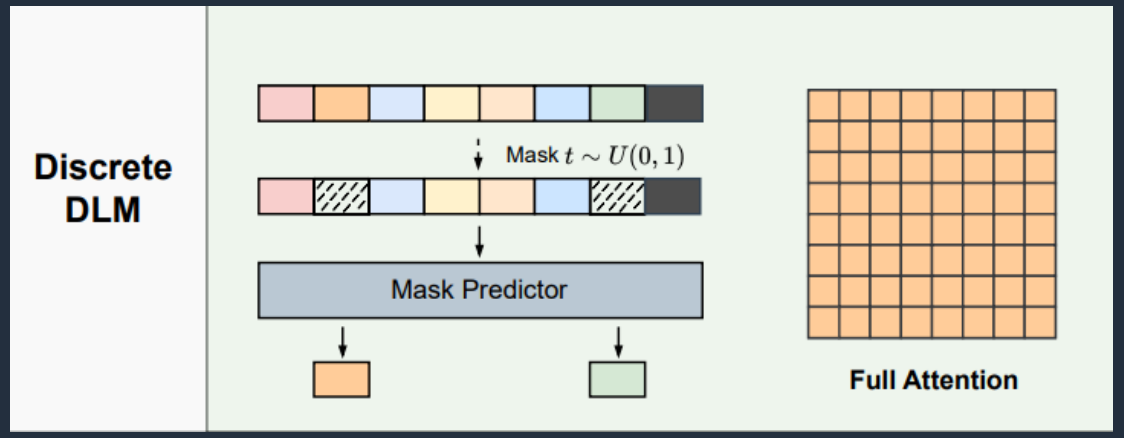
Evolution of Discrete Diffusion Language Models

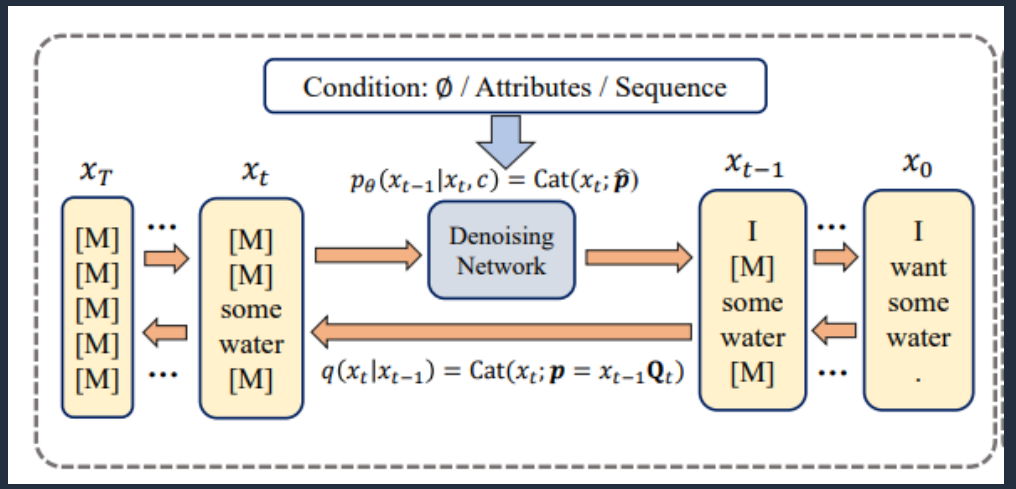


Yu et al., "Discrete Diffusion in Large Language and Multimodal Models: A Survey", 2025.

How can discrete data be modeled as a diffusion process?

- Diffusion in continuous data: add Gaussian noise step-by-step.
- The challenge with discrete data: Text tokens, categorical pixels, or symbolic data are non-continuous
- **Key idea**: instead of Gaussian noise, use **Markov transition matrices** to corrupt data into noise.





D3PM: Diffusion as Transitions in Discrete Space

- Random variables with K categories: $\mathrm{x_t} \in 1, \ldots . K$
- Define the forward transition probabilities by matrices:

$$[Q_t]_{ij} = q(x_t = j \mid x_{t-1} = i\ )$$

- Extend to one-hot version of x

$$q(x_t \mid x_{t-1}) = Cat(x_t; p = x_{t-1}Q_t)$$

Categorical distribution over x given p

e.g. $\mathrm{K} = 3,\ p = [0.1,0.7,0.2] \rightarrow sample\ with\ this\ probability$

D3PM: Diffusion as Transitions in Discrete Space

- Forward marginal

$$q(x_t \mid x_0) = Cat(x_t; p = x_0 \overline{Q_t}) \quad where \quad \overline{Q_t} = Q_1 Q_2 \dots Q_t$$

- Forward posterior

$$q(x_{t-1} \mid x_t, x_0) = \frac{q(x_t \mid x_{t-1}, x_0) q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)}$$

$$= Cat\left(x_{t-1}; p = \frac{x_t Q_t^T \odot x_0 \overline{Q_{t-1}}}{x_0 \overline{Q_t} x_t^T}\right)$$

$$(x_{t-1} Q_t) x_t^T = x_{t-1} (Q_t x_t^T)$$

$$= (x_t \, Q_t^T) x_{t-1}^T$$

Austin et al., "Structured Denoising Diffusion Models in Discrete State-Spaces", NeurIPS 2021.

D3PM: Diffusion as Transitions in Discrete Space

- In continuous diffusion, the forward process is designed so that after many steps the distribution converges to a pure Gaussian.
- Then, what should the transition in discrete diffusion satisfy?

1. **Row stochasticity (probability conservation)**

   : Each row of Q_t must sum to 1.

2. **Known stationary distribution**

   : The product $Q_1 Q_2 \dots Q_T$ must converge to a fixed distribution as $t \to T$.

Austin et al., “Structured Denoising Diffusion Models in Discrete State-Spaces”, NeurIPS 2021.

D3PM: Diffusion as Transitions in Discrete Space

- Choice of Markov transition matrices

1. Uniform

$$[\boldsymbol{Q}_t]_{ij} = \begin{cases} 1 - \frac{K-1}{K}\beta_t & \text{if} \quad i = j \\ \frac{1}{K}\beta_t & \text{if} \quad i \neq j \end{cases}, \quad = (1-\beta_t)I + \beta_t \mathbb{1}\mathbb{1}^T/K$$

➡ **Stationary is uniform**

2. Absorbing state

$$[\boldsymbol{Q}_t]_{ij} = \begin{cases} 1 & \text{if} \quad i = j = m \\ 1-\beta_t & \text{if} \quad i = j \neq m \\ \beta_t & \text{if} \quad j = m, i \neq m \end{cases} \quad = (1-\beta_t)I + \beta_t \mathbb{1} e_m^T$$

➡ **Stationary is [MASK]**

3. Discretized Gaussian

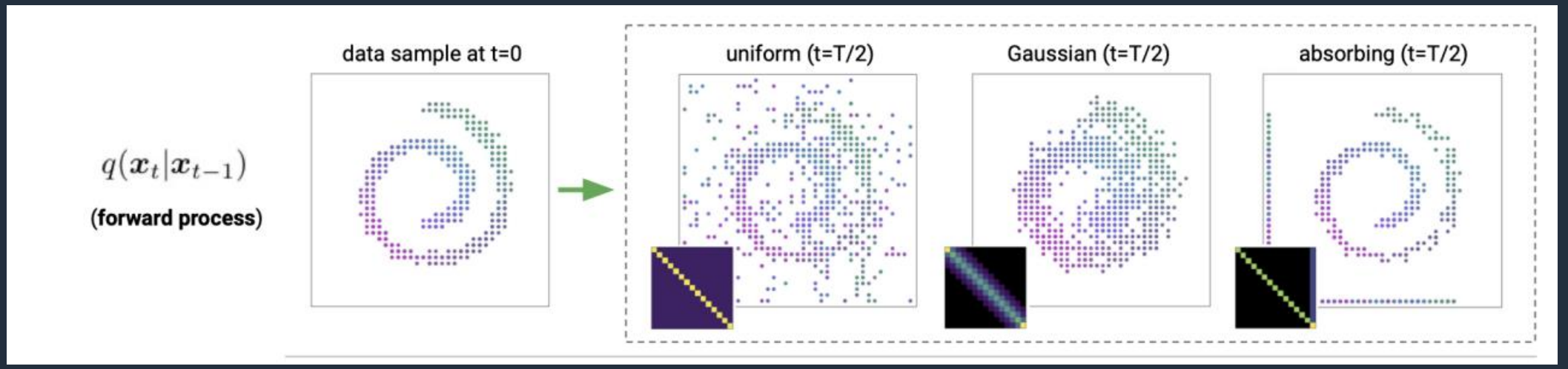
$$[\boldsymbol{Q}_t]_{ij} = \begin{cases} \frac{\exp\left(-\frac{4|i-j|^2}{(K-1)^2\beta_t}\right)}{\sum_{n=-(K-1)}^{K-1}\exp\left(-\frac{4n^2}{(K-1)^2\beta_t}\right)} & \text{if} \quad i \neq j \\ 1 - \sum_{l=0, l\neq i}^{K-1}[\boldsymbol{Q}_t]_{il} & \text{if} \quad i = j \end{cases}$$

➡ **Stationary is uniform**

Austin et al., "Structured Denoising Diffusion Models in Discrete State-Spaces", NeurIPS 2021.

D3PM: Diffusion as Transitions in Discrete Space

- Choice of Markov transition matrices



D3PM: Diffusion as Transitions in Discrete Space

- Reverse process is formulated with x_0–parameterization

  -> Neural network predicts the logits of $\tilde{p}_\theta(\widetilde{x_0}|x_t)$

  -> $p_\theta(x_{t-1}|x_t) \propto \sum_{\widetilde{x_0}} q(x_{t-1}, x_t\,|\widetilde{x_0})\tilde{p}_\theta(\widetilde{x_0}|x_t)$

Recap: D3PM

$$q(x_{t-1} \mid x_t, x_0) = \frac{q(x_t \mid x_{t-1}, x_0)q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)} = Cat\left(x_{t-1}; p = \frac{x_t Q_t^T \odot x_0 \overline{Q_{t-1}}}{x_0 \overline{Q_t} x_t^T}\right)$$

- Sampling process example

$x_t = [0\ 1\ 0]$

$$Q_t = \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.1 & 0.6 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{bmatrix}, \quad Q_{t-1} = \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.2 & 0.5 & 0.3 \\ 0.25 & 0.25 & 0.5 \end{bmatrix}$$

$\tilde{p}_\theta(\widetilde{x_0}|x_t) = [0.1\ 0.6\ 0.3]$

$q(x_{t-1}, x_t\,|1) = [0.16, 0.06, 0.03]$
$q(x_{t-1}, x_t\,|2) = [0.04, 0.30, 0.09]$
$q(x_{t-1}, x_t\,|3) = [0.05, 0.15, 0.15]$

→

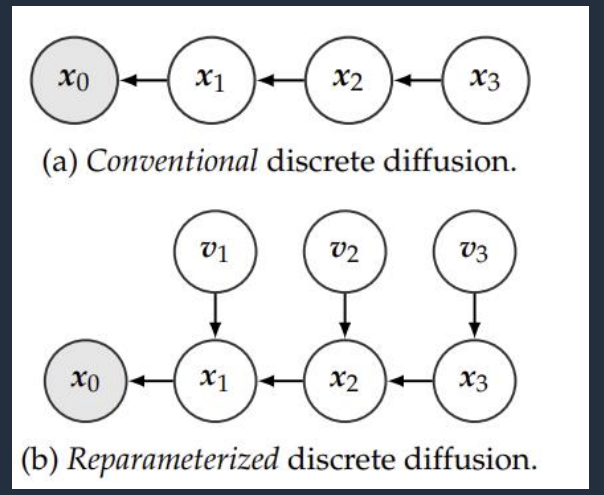
$p_\theta(x_{t-1}|x_t)$
$= 0.1\ [0.16, 0.06, 0.03] + \ldots$
$= [0.055, 0.231, 0.102]$
-> $[0.1418, 0.5954, 0.2629]$

Austin et al., "Structured Denoising Diffusion Models in Discrete State-Spaces", NeurIPS 2021.

RDM: Reparameterized Discrete Diffusion Model

- Backward process in D3PM is quite expensive: summing up over all possible x_0s

$$\mathrm{p}_\theta(x_{t-1}|x_t) \propto \sum_{\widetilde{x_0}} q(x_{t-1}, x_t \,|\widetilde{x_0})\tilde{p}_\theta(\widetilde{x_0}|x_t)$$



- Reparameterize the backward process into a **routing** mechanism:
  - Each token is either denoised (to ground truth) or reset to noise.
  - Introduces latent routing variables controlling token-level decisions.

RDM: Reparameterized Discrete Diffusion Model

- Let the forward transition be:

$$q(x_t \mid x_{t-1}) = \beta_t x_{t-1} + (1-\beta_t) q_{noise}$$

- We can then derive the conditional backward transition as:

$$q(x_{t-1} \mid x_t, x_0) = \begin{cases} \lambda_{t-1}^{(1)} x_t + \left(1 - \lambda_{t-1}^{(1)}\right) q_{noise} & if\ x_t = x_0 \\ \lambda_{t-1}^{(2)} x_0 + \left(1 - \lambda_{t-1}^{(2)}\right) q_{noise}(x_t) & if\ x_t \neq x_0 \end{cases}$$

  - $if\ x_t = x_0$ → Current state is well-formed → Remain at x_t or resets to noise
  - $if\ x_t \neq x_0$ → Current state is noisy → Denoise or remain noisy

RDM: Reparameterized Discrete Diffusion Model

- Reparameterized Sampling

$$b_t = 1_{x_t = x_0}$$

$$v_{t-1}^{(1)} \sim Bernoulli(\lambda_{t-1}^{(1)})$$

$$v_{t-1}^{(2)} \sim Bernoulli(\lambda_{t-1}^{(2)})$$

$$u_t^{(1)} \sim q_{noise}$$

$$u_t^{(2)} \sim q_{noise}(x_t)$$

$$x_{t-1} = b_t[v_{t-1}^{(1)} x_t + (1 - v_{t-1}^{(1)}) u_t^{(1)}] + (1 - b_t)[v_{t-1}^{(2)} x_0 + (1 - v_{t-1}^{(2)}) u_t^{(2)}]$$

-> Simplified training objective & Efficient sampling

Zheng et al., "A Reparameterized Discrete Diffusion Model for Text Generation", COLM 2024.

CTMC: Continuous Time Discrete Denoising Models

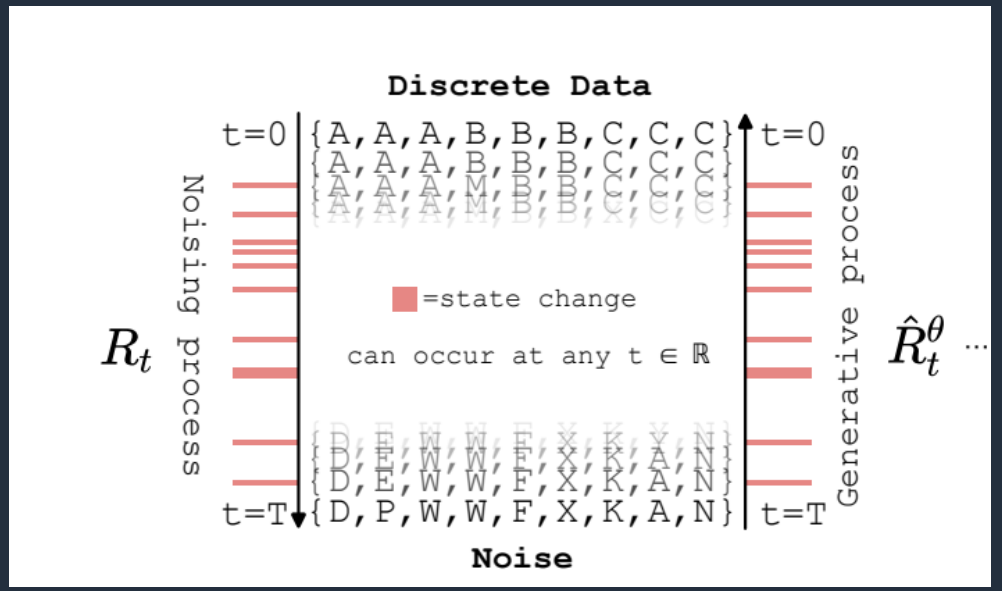
- We have seen the continuous time formulation (ODEs, SDEs) of continuous diffusion models
- Could discrete diffusion be extended to continuous time version?

- **CTMC: Continuous Time Markov Chain**
  - -> Transition can occur at any time t
  - -> Transition is defined by the **rate matrix** $R_t(x, x')$

    instantaneous transition rates

- Infinitesimal transition:

$$q_{t|t-\Delta t}(x' \mid x) = \delta_{x,x'} + R_t(x, x')\Delta t$$



Campbell et al., "A Continuous Time Framework for Discrete Denoising Models", NeurIPS 2022.

CTMC: Continuous Time Discrete Denoising Models

- **CTMC: Continuous Time Markov Chain**

$$q_{t|t-\Delta t}(x'|\, x) = \delta_{x,x'} + R_t(x, x')\Delta t$$

- Reverse transition matrix:

$$\widehat{R_t}(x, x') = R_t(x, x') \sum_{x_0} \frac{q_{t|0}(x'|x_0)}{q_{t|0}(x|x_0)} q_{0|t}(x_0|x)$$

- $q_{0|t}(x_0|x)$: This is intractable -> parametrize by neural net $p_\theta(x_0 \mid x)$

CTMC: Continuous Time Discrete Denoising Models

- Train with continuous time ELBO

$$\mathcal{L}_{\text{vb}}(\theta) = T\,\mathbb{E}_{t\sim\mathcal{U}(0,T)}\,\mathbb{E}_{x\sim q_t}\,\mathbb{E}_{x'\sim r_t(\cdot|x)}\Bigg[\underbrace{\sum_{x''\neq x}\hat{R}_t^\theta(x,x'')}_{\text{outflow from state } x} - \underbrace{Z_t(x)\,\log\hat{R}_t^\theta(x',x)}_{\text{reverse rate matching}}\Bigg] + C$$

$$Z_t(x) = \sum_{x'\neq x} R_t(x,x')$$

$$r_t(x' \mid x) = R_t(x,x')/Z_t(x)$$

- Inference with τ-leaping:

Instead of simulating every single jump event one by one, tau-leaping simulates all jumps over a time interval of length τ at once.

For a current state x_t, number of transitions to x' during [t – τ, t] follows a Poisson distribution:

$$P_{x'} \sim \text{Poisson}(\tau \cdot \hat{R}_t^\theta(x_t, x')).$$

SEDD: Score Entropy Discrete Diffusion models

- Standard diffusion relies on score matching ($\nabla \log p(x)$), well-defined in continuous spaces.

$$dx(t) = [-\frac{1}{2}\beta(t)x(t) - \beta(t)\nabla_x \log q_t(x(t))]dt + \sqrt{\beta(t)}d\bar{\omega}(t)$$

Score Function

- In discrete domains (e.g., text), the gradient is undefined → use concrete score:

CTMC ODE

$$\frac{dp_t}{dt} = Q_t p_t \quad p_0 \approx p_{\text{data}}$$

Euler step

$$p(x_{t+\Delta t} = y | x_t = x) = \delta_{xy} + Q_t(y,x)\Delta t + O(\Delta t^2)$$

Reversal

Concrete Score

$$\frac{dp_{T-t}}{dt} = \overline{Q}_{T-t} p_{T-t} \quad \overline{Q}_t(y,x) = \frac{p_t(y)}{p_t(x)} Q_t(x,y)$$

$$\overline{Q}_t(x,x) = -\sum_{y \neq x} \overline{Q}_t(y,x)$$

Lou et al., "Discrete Diffusion Modeling by Estimating the Ratios of the Data Distribution", ICML 2024.

SEDD: Score Entropy Discrete Diffusion models

- Concrete score matching

$$\mathcal{L}_{\text{CSM}} = \frac{1}{2}\mathbb{E}_{x\sim p_t}\left[\sum_{y\neq x}\left(s_\theta(x_t,t)_y - \frac{p_t(y)}{p_t(x)}\right)^2\right]$$

The ratio must be strictly positive; however, the L2 loss does not guarantee this.

- Denoising score entropy

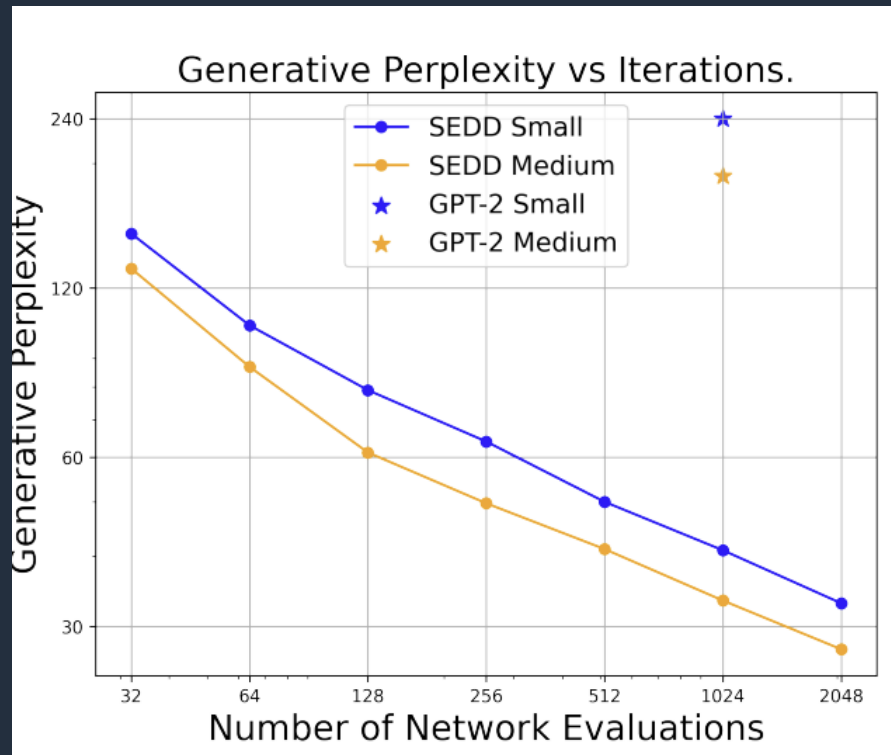
$$\mathcal{L}_{\text{DSE}} = \mathop{\mathbb{E}}_{\substack{x_0\sim p_0\\ x\sim p(\cdot|x_0)}}\left[\sum_{y\neq x} w_{xy}\left(s_\theta(x)_y - \frac{p(y|x_0)}{p(x|x_0)}\log s_\theta(x)_y\right)\right]$$

Enforce positivity -> training is stabilized

Choose $w_{xy} = Q_t(x_t, y)$

Lou et al., “Discrete Diffusion Modeling by Estimating the Ratios of the Data Distribution”, ICML 2024.

SEDD: Score Entropy Discrete Diffusion models



(a) Generative Perplexity (↓) vs. Sampling Iterations.

GPT-2 S	a hiring platform that "includes a fun club meeting place," says petitioner's AQQFredericks. They's the adjacent marijuana-hop. Others have allowed 3B Entertainment
GPT-2 M	misused, whether via Uber, a higher-order reality of quantified impulse or the No Mass Paralysis movement, but the most shamefully universal example is gridlock
SEDD S	As Jeff Romer recently wrote, "The economy has now reached a corner - 64% of household wealth and 80% of wealth goes to credit cards because of government austerity
SEDD M	Wyman worked as a computer science coach before going to work with the U.S. Secret Service in upstate New York in 2010. Without a license, the Secret Service will have to

(b) Generated Text (small models)

Lou et al., "Discrete Diffusion Modeling by Estimating the Ratios of the Data Distribution", ICML 2024.

Masked Diffusion Language Models

MDLM: Simple and Effective Masked Diffusion Language Models

- Assume random variable with K categories (one-hot)
- K-th category is the [MASK] token, m = one-hot vector for this mask

- Define the forward process as the interpolation between clean data and target distribution

$$q(\mathbf{z}_t|\mathbf{x}) = \text{Cat}(\mathbf{z}_t; \alpha_t \mathbf{x} + (1-\alpha_t)\boldsymbol{\pi})$$

Clean data x

Target Distribution $Cat(:, \pi)$

↓

$$q(\mathbf{z}_s|\mathbf{z}_t, \mathbf{x}) = \text{Cat}\left(\mathbf{z}_s; \frac{[\alpha_{t|s}\mathbf{z}_t + (1-\alpha_{t|s})\mathbf{1}\boldsymbol{\pi}^\top \mathbf{z}_t] \odot [\alpha_s \mathbf{x} + (1-\alpha_s)\boldsymbol{\pi}]}{\alpha_t \mathbf{z}_t^\top \mathbf{x} + (1-\alpha_t)\mathbf{z}_t^\top \boldsymbol{\pi}}\right)$$

Recap: D3PM

$$q(x_{t-1} \mid x_t, x_0) = \frac{q(x_t \mid x_{t-1}, x_0) q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)} = Cat\left(x_{t-1}; p = \frac{x_t Q_t^T \odot x_0 \overline{Q_{t-1}}}{x_0 \overline{Q_t} x_t^T}\right)$$

Sahoo et al., "Simple and Effective Masked Diffusion Language Models", NeurIPS 2024.

MDLM: Simple and Effective Masked Diffusion Language Models

- **Masked Diffusion = Absorbing state diffusion**

  Target Distribution $Cat(:,\pi) \rightarrow \pi = m$

- Marginal of the forward

$$q(\mathbf{z}_t|\mathbf{x}) = \alpha_t \mathbf{x} + (1-\alpha_t)\mathbf{m}$$

- Posterior

$$q(\mathbf{z}_s|\mathbf{z}_t,\mathbf{x}) = \begin{cases} \mathrm{Cat}(\mathbf{z}_s;\mathbf{z}_t) & \mathbf{z}_t \neq \mathbf{m}, \\ \mathrm{Cat}\left(\mathbf{z}_s;\frac{(1-\alpha_s)\mathbf{m}+(\alpha_s-\alpha_t)\mathbf{x}}{1-\alpha_t}\right) & \mathbf{z}_t = \mathbf{m}. \end{cases}$$

— Not masked yet

— Interpolate between mask and the true data

Sahoo et al., "Simple and Effective Masked Diffusion Language Models", NeurIPS 2024.

MDLM: Simple and Effective Masked Diffusion Language Models

- True posterior: is unknown (as usual)

$$q(\mathbf{z}_s|\mathbf{z}_t,\mathbf{x}) = \begin{cases} \mathrm{Cat}(\mathbf{z}_s;\mathbf{z}_t) & \mathbf{z}_t \neq \mathbf{m}, \\ \mathrm{Cat}\left(\mathbf{z}_s;\frac{(1-\alpha_s)\mathbf{m}+(\alpha_s-\alpha_t)\mathbf{x}}{1-\alpha_t}\right) & \mathbf{z}_t = \mathbf{m}. \end{cases}$$

- Approximate with the model

$$p_\theta(\mathbf{z}_s|\mathbf{z}_t) = \begin{cases} \mathrm{Cat}(\mathbf{z}_s;\mathbf{z}_t), & \mathbf{z}_t \neq \mathbf{m}, \\ \mathrm{Cat}\left(\mathbf{z}_s;\frac{(1-\alpha_s)\mathbf{m}+(\alpha_s-\alpha_t)\mathbf{x}_\theta(\mathbf{z}_t,t)}{1-\alpha_t}\right). & \mathbf{z}_t = \mathbf{m}. \end{cases}$$

MDLM: Simple and Effective Masked Diffusion Language Models

- Requirements for the model

1. **Zero Masking Probabilities**

   By definition, $< x, m > = 0$

   $\longrightarrow$ $< x_\theta(z_t, t), m > = 0$ $\longrightarrow$ Model should not predict <MASK> as output

2. **Carry-Over Unmasking**

   If we have already unmasked the token, the model should not alter it

   $\longrightarrow$ $x_\theta(z_t, t) = z_t$

MDLM: Simple and Effective Masked Diffusion Language Models

- Loss function

Original ELBO

$$\mathbb{E}_q \Big[\underbrace{\log p_\theta(\mathbf{x}|\mathbf{z}_{t(0)})}_{\mathcal{L}_{\text{recons}}} - \underbrace{\sum_{i=1}^{T} D_{\text{KL}}[q(\mathbf{z}_{s(i)}|\mathbf{z}_{t(i)},\mathbf{x}) \| p_\theta(\mathbf{z}_{s(i)}|\mathbf{z}_{t(i)})]}_{\mathcal{L}_{\text{diffusion}}} \Big] \quad \underbrace{- D_{\text{KL}}[q(\mathbf{z}_{t(T)}|\mathbf{x}) \| p_\theta(\mathbf{z}_{t(T)})]}_{\mathcal{L}_{\text{prior}}}$$

↓

$$q(\mathbf{z}_t|\mathbf{x}) = \text{Cat}(\mathbf{z}_t; \alpha_t \mathbf{x} + (1-\alpha_t)\boldsymbol{\pi})$$

D3PM loss

$$\mathcal{L}_{\text{diffusion}} = \sum_{i=1}^{T} \mathbb{E}_q [\text{D}_{\text{KL}}(q(\mathbf{z}_{s(i)}|\mathbf{z}_{t(i)},\mathbf{x}) \| p_\theta(\mathbf{z}_{s(i)}|\mathbf{z}_{t(i)}))] = \sum_{i=1}^{T} \mathbb{E}_q \Big[\frac{\alpha_s - \alpha_t}{1-\alpha_t} \log \frac{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t,t),\mathbf{m} \rangle + (1-\alpha_t)}{(1-\alpha_t)\langle \mathbf{x}_\theta(\mathbf{z}_t,t),\mathbf{x} \rangle} + \frac{1-\alpha_s}{1-\alpha_t} \log \frac{(1-\alpha_s)(\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t,t),\mathbf{m} \rangle + (1-\alpha_t))}{(1-\alpha_t)(\alpha_s \langle \mathbf{x}_\theta(\mathbf{z}_t,t),\mathbf{m} \rangle + (1-\alpha_s))} \Big] \langle \mathbf{z}_t,\mathbf{m} \rangle.$$

Sahoo et al., "Simple and Effective Masked Diffusion Language Models", NeurIPS 2024.

MDLM: Simple and Effective Masked Diffusion Language Models

- Loss function

D3PM loss

$$\mathcal{L}_{\text{diffusion}} = \sum_{i=1}^{T} \mathbb{E}_q [\mathrm{D}_{\mathrm{KL}}(q(\mathbf{z}_{s(i)}|\mathbf{z}_{t(i)},\mathbf{x}) \| p_\theta(\mathbf{z}_{s(i)}|\mathbf{z}_{t(i)}))] = \sum_{i=1}^{T} \mathbb{E}_q \left[\frac{\alpha_s - \alpha_t}{1-\alpha_t} \log \frac{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t,t),\mathbf{m}\rangle + (1-\alpha_t)}{(1-\alpha_t)\langle \mathbf{x}_\theta(\mathbf{z}_t,t),\mathbf{x}\rangle} + \frac{1-\alpha_s}{1-\alpha_t} \log \frac{(1-\alpha_s)(\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t,t),\mathbf{m}\rangle + (1-\alpha_t))}{(1-\alpha_t)(\alpha_s \langle \mathbf{x}_\theta(\mathbf{z}_t,t),\mathbf{m}\rangle + (1-\alpha_s))} \right] \langle \mathbf{z}_t,\mathbf{m}\rangle.$$

↓ $< x_\theta(z_t, t), m > = 0$

MDLM loss (Rao-Blackwellization)

$$\mathcal{L}_{\text{diffusion}} = \sum_{i=1}^{T} \mathbb{E}_q [\mathrm{D}_{\mathrm{KL}}(q(\mathbf{z}_{s(i)}|\mathbf{z}_{t(i)},\mathbf{x}) \| p_\theta(\mathbf{z}_{s(i)}|\mathbf{z}_{t(i)}))] = \sum_{i=1}^{T} \mathbb{E}_q \left[\frac{\alpha_{t(i)} - \alpha_{s(i)}}{1-\alpha_{t(i)}} \log \langle \mathbf{x}_\theta(\mathbf{z}_{t(i)}),\mathbf{x}\rangle \right].$$

Sahoo et al., "Simple and Effective Masked Diffusion Language Models", NeurIPS 2024.

MDLM: Simple and Effective Masked Diffusion Language Models

- Continuous time formulation

  Sending $T \to \infty$

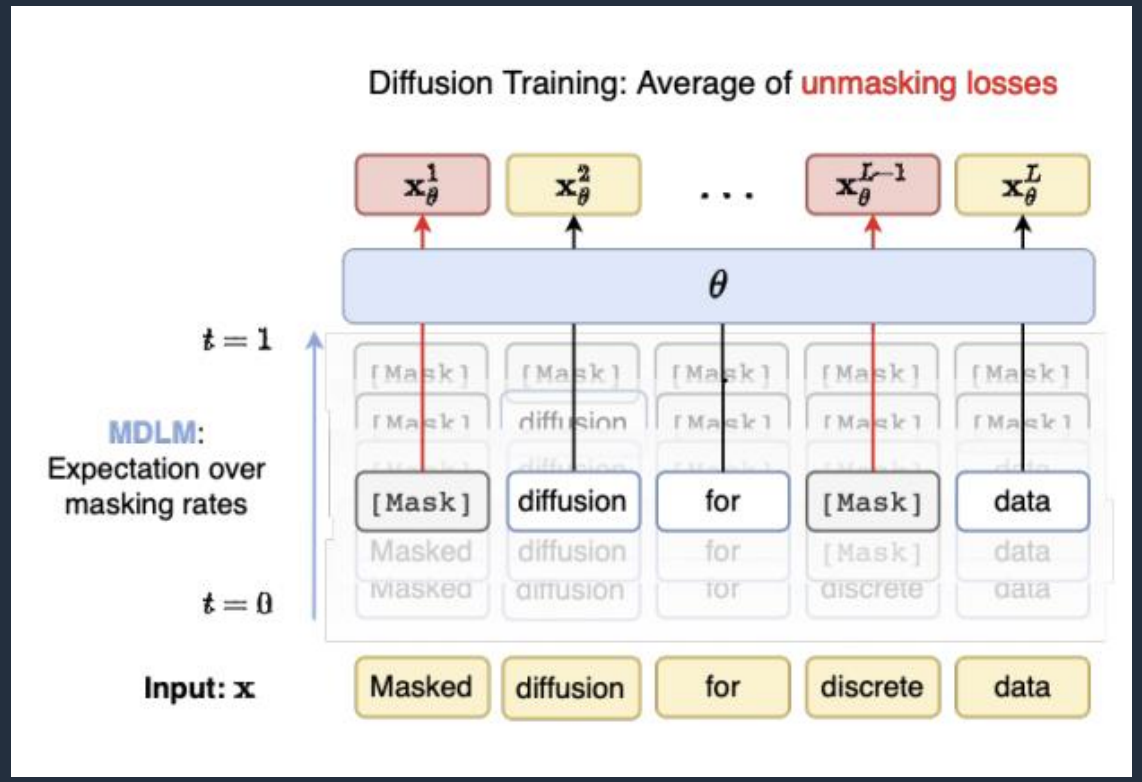
$$\mathcal{L}^{\infty}_{\text{diffusion}} = \mathbb{E}_q \int_{t=0}^{t=1} \frac{\alpha'_t}{1-\alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \mathrm{d}t$$

- Masked diffusion language model

  Sequence of $x^{1:L}$

$$\mathcal{L}^{\infty}_{\text{diffusion}} = \mathbb{E}_q \int_{t=0}^{t=1} \frac{\alpha'_t}{1-\alpha_t} \sum_{\ell} \log \langle \mathbf{x}^{\ell}_\theta(\mathbf{z}_t), \mathbf{x}^{\ell} \rangle \mathrm{d}t$$

Weighted average of MLM loss!



RADD: Reparameterized Absorbing Discrete Diffusion

- Recall: Continuous time formulation of discrete diffusion model

CTMC ODE

$$\frac{dp_t}{dt} = Q_t p_t \quad p_0 \approx p_{\text{data}}$$

Euler step

$$p(x_{t+\Delta t} = y | x_t = x) = \delta_{xy} + Q_t(y,x)\Delta t + O(\Delta t^2)$$

Reversal

$$\frac{dp_{T-t}}{dt} = \overline{Q}_{T-t} p_{T-t} \quad \overline{Q}_t(y,x) = \frac{p_t(y)}{p_t(x)} Q_t(x,y)$$

$$\overline{Q}_t(x,x) = -\sum_{y \neq x} \overline{Q}_t(y,x)$$

Concrete Score

For multi-dimensional setting (length d), score becomes (considering the cases when only 1 tokens are changed):

→

$$\boldsymbol{s}_\theta(\boldsymbol{x}_t, t)_{\hat{\boldsymbol{x}}_t} = \boldsymbol{s}_\theta\left(x_t^1 \dots x_t^i \dots x_t^d, t\right)[i, \widehat{x}_t^i] \approx \frac{p_t\left(x_t^1 \dots \widehat{x}_t^i \dots x_t^d\right)}{p_t\left(x_t^1 \dots x_t^i \dots x_t^d\right)},$$

RADD: Reparameterized Absorbing Discrete Diffusion

$$s_\theta\left(\boldsymbol{x}_t, t\right)_{\hat{\boldsymbol{x}}_t} = s_\theta\left(x_t^1 \ldots x_t^i \ldots x_t^d, t\right)[i, \widehat{x}_t^i] \approx \frac{p_t\left(x_t^1 \ldots \widehat{x}_t^i \ldots x_t^d\right)}{p_t\left(x_t^1 \ldots x_t^i \ldots x_t^d\right)},$$

⟶ The reverse transition rate matrix then becomes:

$$\begin{aligned}\tilde{Q}_t\left(x_t^1 \ldots x_t^i \ldots x_t^d, x_t^1 \ldots \widehat{x}_t^i \ldots x_t^d\right) &= Q_t^{\text{tok}}\left(\widehat{x}_t^i, x_t^i\right) \frac{p_t\left(x_t^1 \ldots \widehat{x}_t^i \ldots x_t^d\right)}{p_t\left(x_t^1 \ldots x_t^i \ldots x_t^d\right)} \\ &\approx Q_t^{\text{tok}}\left(\widehat{x}_t^i, x_t^i\right) s_\theta\left(x_t^1 \ldots x_t^i \ldots x_t^d, t\right)[i, \widehat{x}_t^i].\end{aligned}$$

⟶ Note that for masked diffusion, only the transition from the masked token to an unmasked token is valid

$$\tilde{Q}_t\left(x_t^1 \ldots x_t^i \ldots x_t^d, x_t^1 \ldots \widehat{x}_t^i \ldots x_t^d\right) = \sigma(t) Q^{\text{absorb}}\left(\widehat{x}_t^i, x_t^i\right) \frac{p_t\left(x_t^1 \ldots \widehat{x}_t^i \ldots x_t^d\right)}{p_t\left(x_t^1 \ldots x_t^i \ldots x_t^d\right)} = 0, \quad \begin{array}{l} x_t^i \neq [\mathbf{M}] \\ \\ \hat{x}_t^i \neq x_t^i \end{array}$$

Ou et al., "YOUR ABSORBING DISCRETE DIFFUSION SECRETLY MODELS THE CONDITIONAL DISTRIBUTIONS OF CLEAN DATA", ICLR 2025.

RADD: Reparameterized Absorbing Discrete Diffusion

→ Then, we can express the concrete score a time-independent conditional distribution at time zero multiplied by an analytic time-dependent term

$$\underbrace{\frac{p_t\left(x_t^1 \dots \hat{x}_t^i \dots x_t^d\right)}{p_t\left(x_t^1 \dots x_t^i \dots x_t^d\right)}}_{\textit{concrete score}} = \underbrace{\frac{e^{-\bar{\sigma}(t)}}{1-e^{-\bar{\sigma}(t)}}}_{\textit{scalar}} \underbrace{p_0(\hat{x}_t^i | \boldsymbol{x}_t^{UM})}_{\textit{clean data distribution}}$$

$$x_t^i = [\boldsymbol{M}] \textit{ and } \hat{x}_t^i \neq [\boldsymbol{M}]$$

$\boldsymbol{x}_t^{UM}$ *is the vector consists of all unmasked tokens of* $\boldsymbol{x}_t$

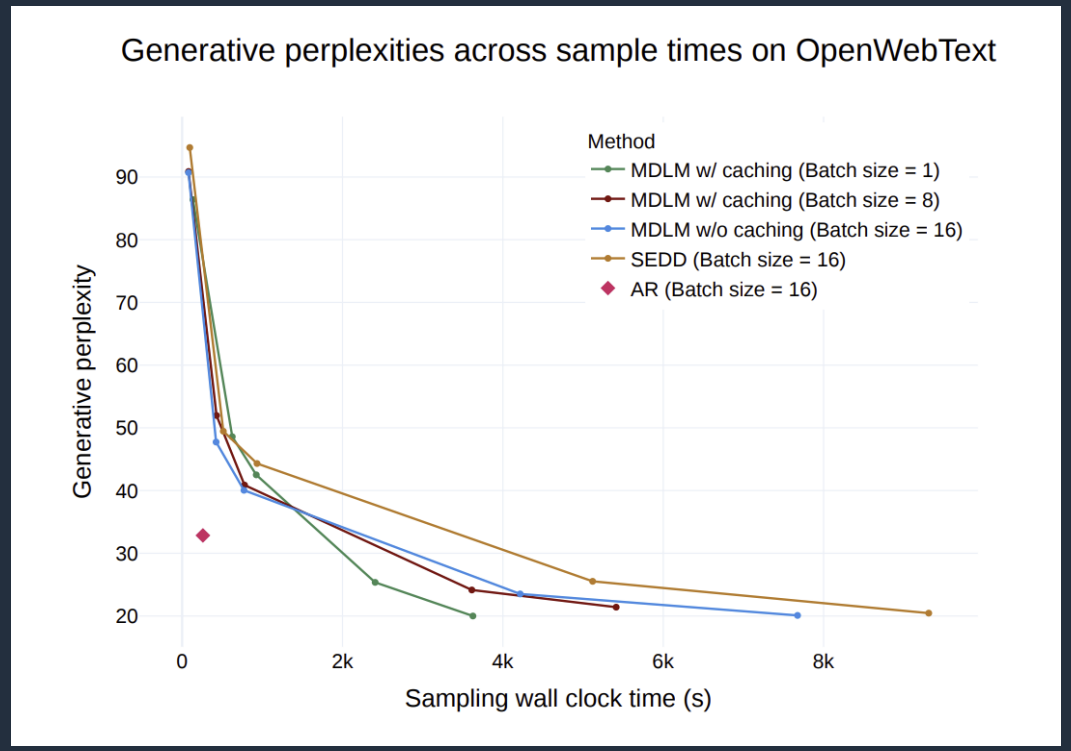
→ Removes time dependency

→ Enables caching

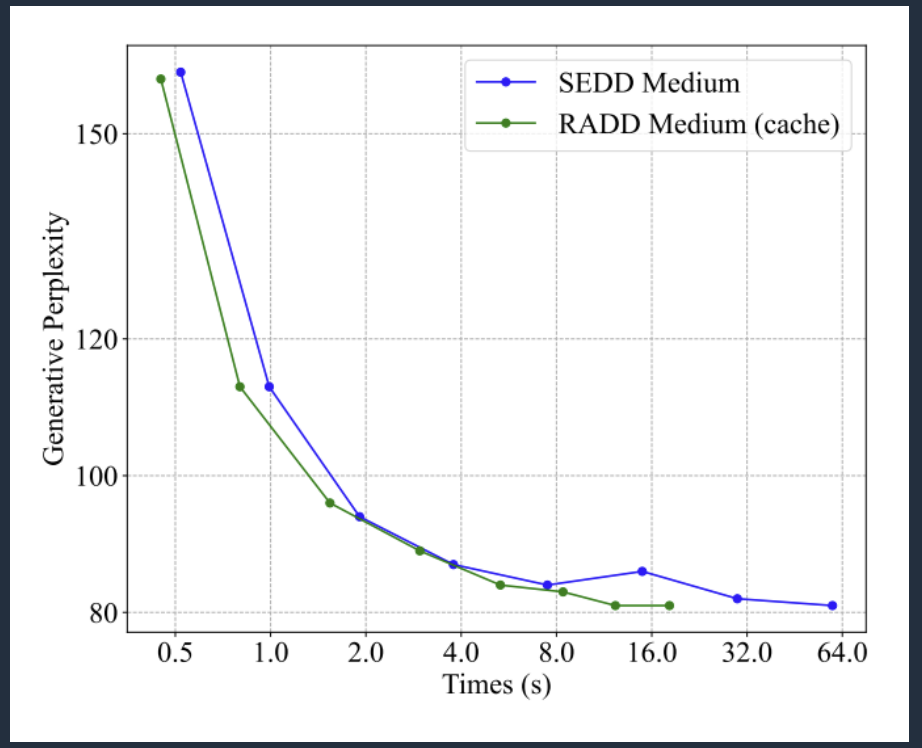
Ou et al., "YOUR ABSORBING DISCRETE DIFFUSION SECRETLY MODELS THE CONDITIONAL DISTRIBUTIONS OF CLEAN DATA", ICLR 2025.

RADD: Reparameterized Absorbing Discrete Diffusion

MDLM



RADD

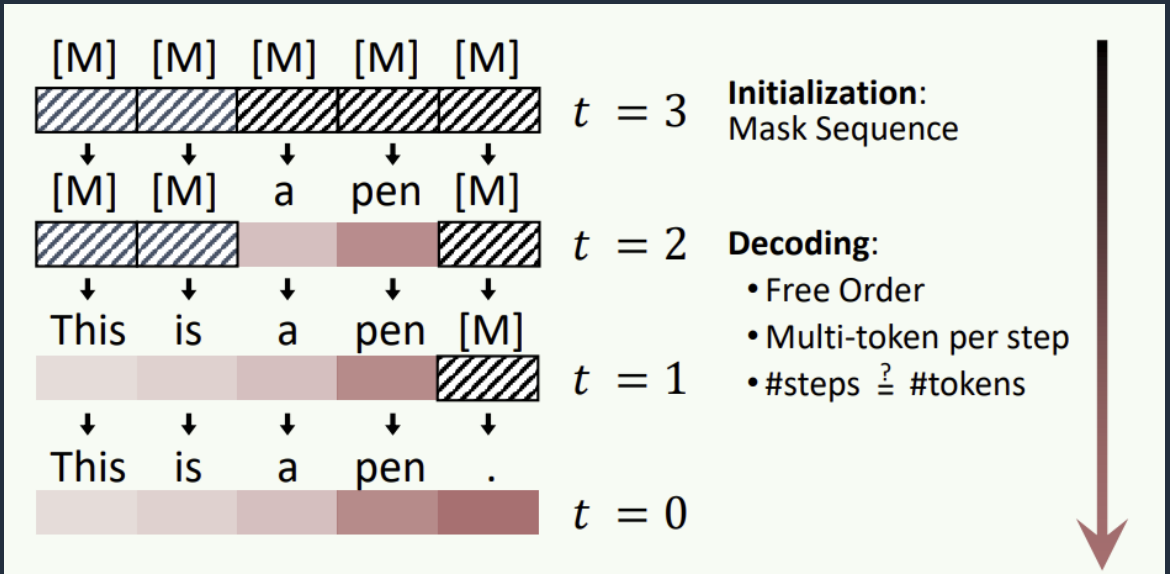


Ou et al., "YOUR ABSORBING DISCRETE DIFFUSION SECRETLY MODELS THE CONDITIONAL DISTRIBUTIONS OF CLEAN DATA", ICLR 2025.

Unmasking
& Token Ordering

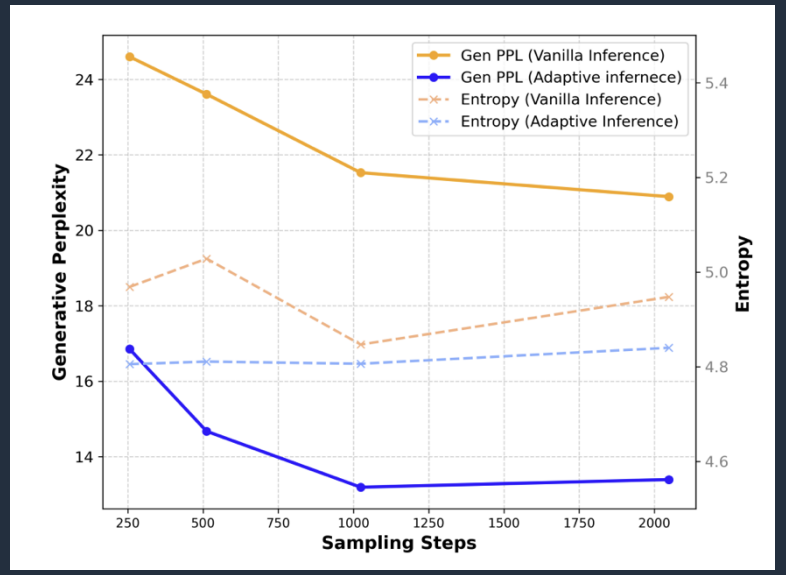
Unmasking

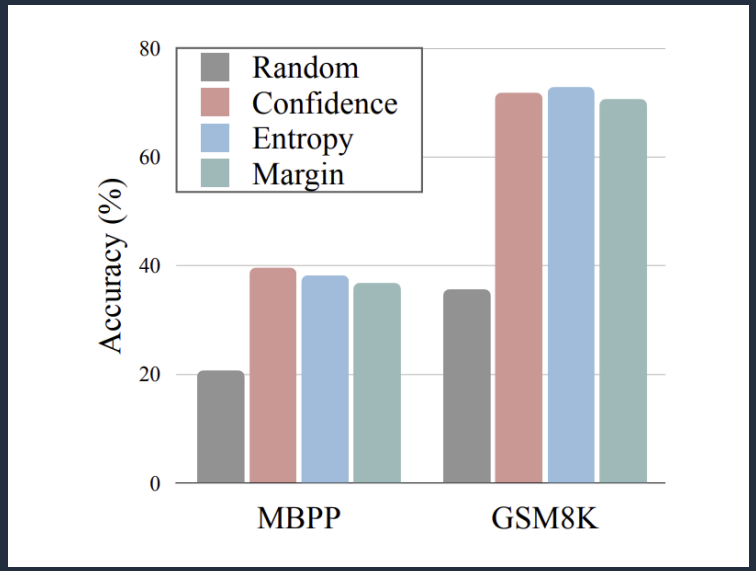
- Discrete diffusion can decode many tokens at once, since it operates over the entire sequence simultaneously.
- However, because unmasked tokens are often unchanged afterward, it is unclear whether unmasking all tokens is always beneficial.



Order Matters

- Recent works in discrete diffusion show that the unmasking strategy has a significant impact on performance..
- This raises the question of **token ordering**—specifically, which tokens should be prioritized and unmasked earlier versus later.



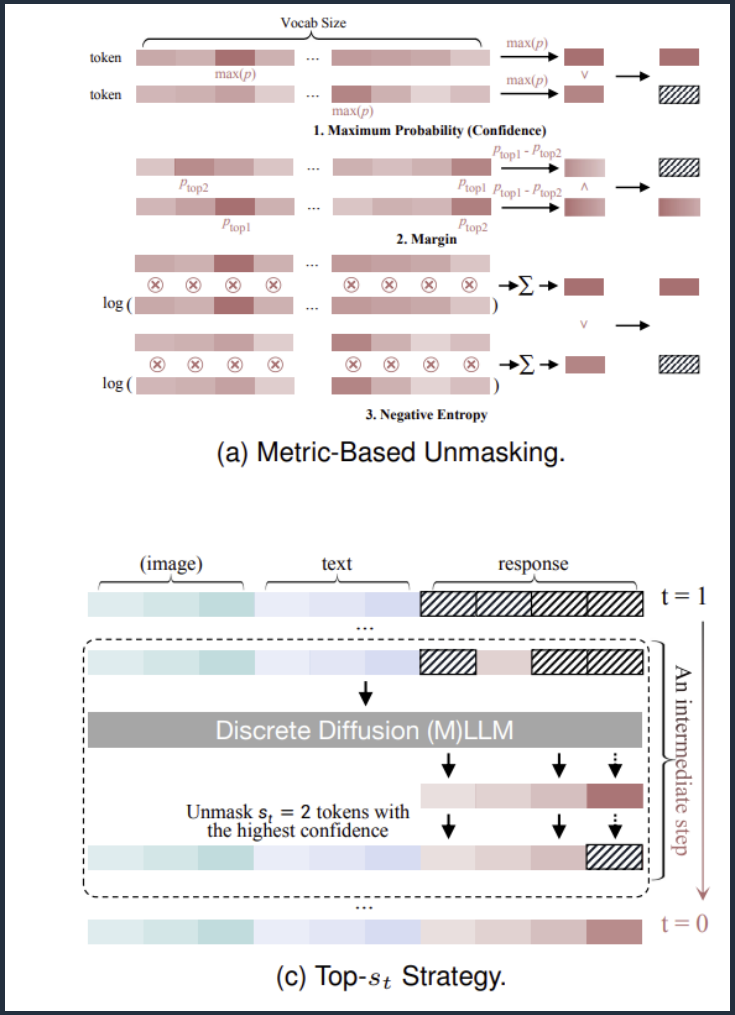


How to unmask?

- Then the questions are:

1. Based on what criteria should we unmask?

2. And how much should be unmasked?



Criteria for Unmasking

1. Random

- Randomly select K_t positions to unmask at time t

2. Confidence (SMDM, 2025)

- Select K_t positions based on confidence

$$\text{Confidence: } l = \underset{l' \in M}{\text{argmax}} \left[\max_{x^{l'}} p^{\theta}(x^{l'} | x^{\bar{M}}) \right]$$

3. Entropy (Dream, 2025)

- Select K_t positions based on entropy

$$\text{Entropy: } l = \underset{l' \in M}{\text{argmin}} \left[H(X^{l'} | X^{\bar{M}} = x^{\bar{M}}) \right]$$

4. Margin (Kim et al., 2025)

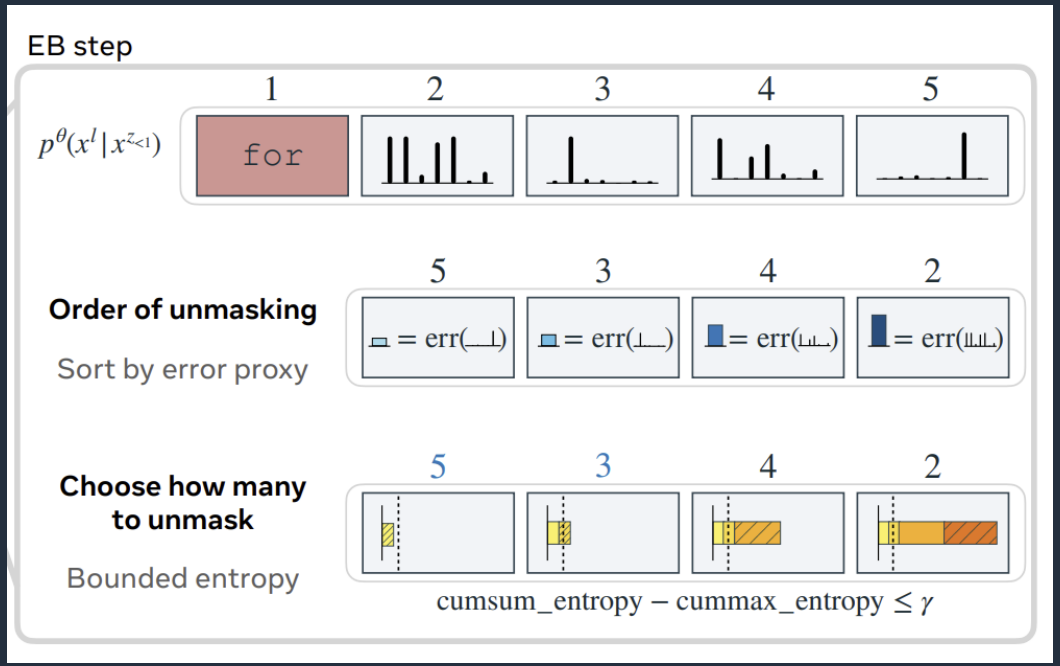
- Select K_t positions based on confidence margin

$$\text{Margin: } l = \underset{l' \in M}{\text{argmax}} \left[p^{\theta}(X^{l'} = y_1 | x^{\bar{M}}) - p^{\theta}(X^{l'} = y_2 | x^{\bar{M}}) \right]$$

Ben-Hamu et al., "Accelerated Sampling from Masked Diffusion Models via Entropy Bounded Unmasking", 2025.

Selecting K

1. **Top-K**
   - Unmask a fixed number of tokens at each step
2. **Confidence threshold (Dimple, 2025)**
   - Dynamically unmask all tokens above a fixed confidence threshold
3. **Adaptive (EB-Sampler, 2025)**
   - Adjust the number of tokens unmasked per step based on an entropy bound, unmasking more when the model is confident



Remasking

- Then, beyond unmasking newly generated tokens, we may also wish to revisit previously unmasked tokens.

MDLM

$$q(\mathbf{z}_s|\mathbf{z}_t,\mathbf{x}) = \begin{cases} \mathrm{Cat}(\mathbf{z}_s;\mathbf{z}_t) & \mathbf{z}_t \neq \mathbf{m}, \\ \mathrm{Cat}\left(\mathbf{z}_s;\frac{(1-\alpha_s)\mathbf{m}+(\alpha_s-\alpha_t)\mathbf{x}}{1-\alpha_t}\right) & \mathbf{z}_t = \mathbf{m}. \end{cases}$$

ReMDM

$$q_\sigma(\mathbf{z}_s \mid \mathbf{z}_t, \mathbf{x}) = \begin{cases} \mathrm{Cat}(\mathbf{z}_s; (1-\sigma_t)\mathbf{x} + \sigma_t \boldsymbol{m}), & \mathbf{z}_t \neq \boldsymbol{m} \\ \mathrm{Cat}(\mathbf{z}_s; \frac{\alpha_s-(1-\sigma_t)\alpha_t}{1-\alpha_t}\mathbf{x} + \frac{1-\alpha_s-\sigma_t\alpha_t}{1-\alpha_t}\boldsymbol{m}), & \mathbf{z}_t = \boldsymbol{m}. \end{cases}$$

Wang et al., "Remasking Discrete Diffusion Models with Inference-Time Scaling", 2025.

Next week

Week 8
Diffusion Language Models III

- AR VS. NAR
- AR-like formulation in DLMs
- Block-wise diffusion

Week 9
Diffusion Language Models IV

- Recent advances in DLM
- Scaling, Reasoning
- KV-caching, distillation

More on this...

- Discrete diffusion modeling
  - Austin et al., “Structured Denoising Diffusion Models in Discrete State-Spaces”, NeurIPS 2021.
  - Hoogebum et al., “Autoregressive Diffusion Models”, ICLR 2022.

- CTMC / Concrete score matching / Ratio matching
  - Campbell et al., “A Continuous Time Framework for Discrete Denoising Models”, NeurIPS 2022.
  - Meng et al., “Concrete Score Matching: Generalized Score Matching for Discrete Data”, NeurIPS 2022.
  - Lou et al., “Discrete Diffusion Modeling by Estimating the Ratios of the Data Distribution”, ICML 2024.
  - Ou et al., “YOUR ABSORBING DISCRETE DIFFUSION SECRETLY MODELS THE CONDITIONAL DISTRIBUTIONS OF CLEAN DATA”, ICLR 2025.
  - Haxholi et al., “EFFICIENT PERPLEXITY BOUND AND RATIO MATCHING IN DISCRETE DIFFUSION LANGUAGE MODELS ”, ICLR 2025.

More on this...

- Masked diffusion language modeling
  - He et al., “DiffusionBERT: Improving Generative Masked Language Models with Diffusion Models”, ACL 2023.
  - Zhou et al., “Diffusion-NAT: Self-Prompting Discrete Diffusion for Non-Autoregressive Text Generation”, EACL 2024.
  - Zheng et al., “A Reparameterized Discrete Diffusion Model for Text Generation”, COLM 2024.
  - Sahoo et al., “Simple and Effective Masked Diffusion Language Models”, NeurIPS 2024.
  - Nie et al., “Scaling up Masked Diffusion Models on Text”, ICLR 2025.
  - Nie et al., “Large Language Diffusion Models”, 2025.
  - Gong et al., “Scaling Diffusion Language Models via Adaptation from Autoregressive Models”, ICLR 2025.
- Uniform state diffusion language modeling
  - Schiff et al., “Simple Guidance Mechanisms for Discrete Diffusion Models”, ICLR 2025.
  - Sahoo et al., “The Diffusion Duality”, ICML 2025.
- Unmasking & Token ordering
  - Kim et al., “Train for the Worst, Plan for the Best: Understanding Token Ordering in Masked Diffusions”, ICML 2025.
  - Ben-Hamu et al., “Accelerated Sampling from Masked Diffusion Models via Entropy Bounded Unmasking”, 2025.

Thank you!

Woojin Kim

AIDAS LAB
Seoul National University
wjk9904@snu.ac.kr

AIDAS Lab